Phone Test: Easy Steps

No cable, no router. Just a phone hotspot and this laptop.

What you need

- This laptop.
- One phone to make a hotspot (yours).
- The phones to test: yours and friends' (ask them first). Best to have: an **iPhone with Safari**, an **Android with Chrome**, and **Firefox** on Android if possible.
- About 2 minutes per phone.

Part 1: Start (once)

1. On your phone, turn on **Mobile Hotspot**. Mobile data must be on.
2. Connect the **laptop's Wi-Fi** to that hotspot.
3. On the laptop, open the folder `Desktop\cnfinalpaper\hardware_test`.
4. Click the folder's address bar at the top, type `cmd`, press **Enter**. A black window opens.
5. Type `python server.py` and press **Enter**.
6. It prints one or more addresses like `http://192.168.43.12:8000/`. Write down the one that starts with `192.168`, `172.20` or `10.`
7. If Windows asks about the firewall, tick **both** Private and Public, then press **Allow**.
8. Keep the black window open until you finish.

Part 2: Test each phone

1. Connect the phone to the **same hotspot**. (The hotspot phone itself can also be tested; if the page will not open on it, skip it.)
2. Open the browser you want to test (Safari, Chrome or Firefox).
3. Type the address from Part 1, step 6, exactly, including `:8000`. A page called **WebRTC path test** opens.
4. In the box **This device and browser**, type the phone and browser, for example `iPhone 13 Safari` or `Redmi Note 12 Chrome`.
5. In the box **Network setup**, type `hotspot`.
6. Press **Test this device only**. Wait about 10 seconds.
7. The screen shows **Result: HIDES local address** or **Result: REVEALS local address**. It is saved on the laptop automatically.
8. **Reload the page and press the button again**, so each phone is tested **3 times**.

9. Repeat for every phone and every browser.

Part 3: Write down (once per phone)

Phone model	**OS version** (Settings → About)	**Browser and version** (browser → About)	**Result (3 runs)**

Part 4: Finish

1. In the black window press **Ctrl + C**, then close it.
2. A file `results.jsonl` is now in the `hardware_test` folder. It holds only address *types*, never the actual addresses.
3. Send that file and the table above to Claude to add to the paper.

If something goes wrong

Problem	**Fix**
Page does not open on the phone	Check the phone is on the same hotspot. Try the other address the black window printed. Allow Python in the firewall for Public networks.
`python` is not recognised	Close the black window and open it again from the folder; if it still fails, ask Claude.
Result: NO CANDIDATES	Turn mobile data on; the hotspot needs internet.
iPhone hotspot: laptop cannot connect	On the iPhone, keep the Personal Hotspot screen open while connecting.

What this test does and does not show. It shows what each phone browser reveals about its own address. It does not test client isolation, because a phone hotspot has no such setting. Each phone talks only to the laptop and to Google's public STUN server, as in a normal video call.